9.5 Composition of Functions

A Composition of Functions

Any function $y = f(x)$ can be imagined as a machine f who changes the input x into the output y:

$$x \rightarrow \boxed{f} \rightarrow y$$

Let connect now the output of the function machine g as the input of the function machine f corresponding to a different function $z = f(y)$:

$$x \rightarrow \boxed{g} \rightarrow y \rightarrow \boxed{f} \rightarrow z$$

Then, the following expressions can be written:

$$y = g(x),\ z = f(y),\ z = f(g(x))$$

So, we can replace the machines g and f by a new machine $f \circ g$ who changes the input x directly into the output z:

$$x \rightarrow \boxed{f \circ g} \rightarrow z$$

So, the relation

$$(f \circ g)(x) = f(g(x))$$

defines the composition of the functions f and g.

Similarly:

$$(g \circ f)(x) = g(f(x))$$
$$(f \circ f)(x) = f(f(x))$$
$$(g \circ g)(x) = g(g(x))$$

Ex 1. Consider the following functions $f(x) = x + 1$, $g(x) = x^2 - 2$, $h(x) = \sqrt{x} + 3$. Find:

a) $(f \circ g)(0) = f(g(0)) = f(-2) = -1$

b) $(g \circ f)(2) = g(f(2)) = g(3) = 9 - 2 = 7$

c) $(f \circ h)(4) = f(h(4)) = f(5) = 5 + 1 = 6$

d) $(h \circ f)(0) = h(f(0)) = h(1) = 4$

e) $(f \circ f)(2) = f(f(2)) = f(3) = 4$

f) $(g \circ g)(1) = g(g(1)) = g(-1) = -1$

g) $(h \circ h)(1) = h(h(1)) = h(4) = 5$

i) $(f \circ g \circ h)(1) = f(g(h(1))) = f(g(4)) =$
$= f(14)$
$= 14 + 1 = 15$

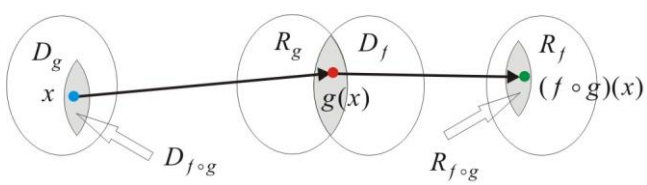
B Domain and Range

The domain of $f \circ g$ is a subset of the domain of g:

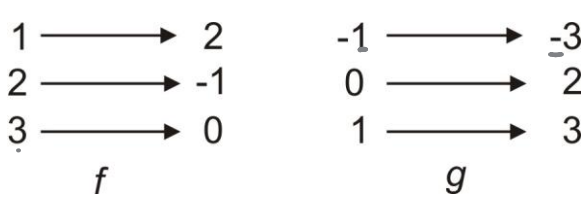
$$D_{f \circ g} = \{x \in D_g \mid g(x) \in D_f\}$$

The domain of $f \circ g$ consists of the numbers x in the domain of g such that $g(x)$ is in the domain of f.

The range of $f \circ g$ is a subset of the range of f.



Ex 2. The functions f and g are given by the following mapping diagrams. Find each composition and then find the domain and the range of the composition.



a) $f \circ g = \{(0,-1), (1,0)\}$

$D_{f \circ g} = \{0, 1\}$ $R_{f \circ g} = \{(-1, 0)\}$

b) $g \circ f = \{(2,-3), (3,2)\}$

$D_{g \circ f} = \{2, 3\}$

$R_{g \circ f} = \{-3, 2\}$

c) $f \circ f = \{(1,-1)\}$

$D_{f \circ f} = \{1\}$

$R_{f \circ f} = \{-1\}$

d) $g \circ g = \{\phi\}$

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Ex 3. Consider the following functions $f(x)=2x-1$, $g(x)=x^2$, $h(x)=\sqrt{x-1}$, and $k(x)=\dfrac{1}{x+1}$. Find the following compositions and for each case state the domain and the range.

a) $(f\circ g)(x) = f(x^2)=2x^2-1$

$$x \xrightarrow{g} y=x^2 \xrightarrow{f} z=2y-1$$

$x\in R$; $y\geq 0$, $y\in R$; $y\geq 0$; $z\geq -1$, $z\in R$

$D_{f\circ g}=\{x\in R\}$

$R_{f\circ g}=\{y\in R / y\geq -1\}$

b) $(k\circ k)(x) = k(k(x)) = k\left(\frac{1}{x+1}\right) = \dfrac{1}{\frac{1}{x+1}+1} = \dfrac{x+1}{x+2}$

$$x \xrightarrow{k} y=\frac{1}{x+1} \xrightarrow{k} z=\frac{1}{y+1}$$

$x\neq -1$; $y\neq 0$, $y\neq -1$; $z\neq 0$, $z\neq 1$; $x\neq -2$

$\therefore$ $D_{k\circ k}=\{x\in R / x\neq -1, -2\}$

$R_{k\circ k}=\{y\in R / y\neq 0, 1\}$

c) $(f\circ g\circ h)(x) = f(g(\sqrt{x-1})) = f(x-1) = 2(x-1)-1 = 2x-3$

$$x \xrightarrow{h} y=\sqrt{x-1} \xrightarrow{g} z=y^2 \xrightarrow{f} w=2z-1$$

$x\geq 1$; $y\geq 0$, $y\in R$; $z\geq 0$, $z\in R$; $w\in R$; $w\geq -1$

$D_{f\circ g\circ h}=\{x\in R\}$

$R_{f\circ g\circ h}=\{y\in R / y\geq -1\}$

d) $(k\circ g\circ f)(x) = k(g(f(x))) = k(g(2x-1)) = k((2x-1)^2) = \dfrac{1}{(2x-1)^2+1}$

$$x \xrightarrow{f} y=2x-1 \xrightarrow{g} z=y^2 \xrightarrow{k} w=\frac{1}{z+1}$$

$x\in R$; $y\in R$, $y\in R$; $z\geq 0$, $z\neq -1$ always.; $w\neq 0$

$\therefore$ $D=\{x\in R\}$

$R=\{y\in R / y\neq 0\}$

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Ex 4. Prove that:

a) $(f \circ f^{-1})(x) = x$

$(f \circ f^{-1})(x)$

$$x \xrightarrow{f} y$$
$$x \xleftarrow{f^{-1}} y$$

$y = f(x)$
$x = f^{-1}(y)$

$x = f^{-1}(f(x))$
$x = f^{-1} \circ f(x)$

$y = f(x)$
$= f(f^{-1}(y))$
$f \circ f^{-1}(y) = y$

b) $(f^{-1} \circ f)(x) = x$

Ex 5. Prove that:

$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

$(f \circ g)^{-1} = g^{-1}(f^{-1})$

$$x \xrightarrow{g} y = g(x) \xrightarrow{f} z = f(y)$$
$$x \xleftarrow{g^{-1}} y \xleftarrow{f^{-1}} z$$

Reading: Nelson Textbook, Pages 545-551
Homework: Nelson Textbook, Page 552: #5aef, 6ace, 7ac, 9, 13, 16

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